Victor **Gaya**

AI Research Engineer | Deep Learning & Foundation Models

Profile

AI researcher with a PhD and 3 years of experience in deep learning applied to satellite imagery. Comfortable working on the full ML pipeline, from data preparation to model training and evaluation. Looking to transition to industry and bring my research background to applied engineering challenges.

Contact

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LinkedIn : Victor Gaya

Research Gate: Victor Gaya

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Languages

French - Native Language

English - Fluent (TOEIC C1, 2022)

Technical Skills

Python, PyTorch, TensorFlow, Git

Docker, Linux, Bash

HuggingFace, scikit-learn, W&B

LaTeX, MS Office

Involvements

- Drone association founder at CEA
- Concert organiser & technical manager
- Student trips association president
- School annual Gala organiser

Other Interests

Music production

Tennis

Mountaineering

Climbing

WORK EXPERIENCE

Jan. 2026 - Today

Research fellow; AI & aerospace — Singapore, Singapore
NUS
National University of Singapore, ECE

- Applying **Foundation Models** and **Generative AI** to analyze complex **satellite imagery**.
- Generalizing learned representations across domains via **data-driven approaches**.
- Supervising **interns** on experimental design and validating **research outcomes**.

Dec. 2022 - Dec. 2025

Junior researcher; AI & aerospace — Arpajon, France
CEA
Research center

- Developed **Deep Learning** for **SAR interferometry** and **3D reconstruction** in industrial contexts.
- Combined **data fusion** and **auto-supervised learning** based on **statistical modeling**.
- Disseminated results at **international conferences** (oral and poster).

Mar. 2022 - Sept. 2022

Machine Learning intern — Toulouse, France
CNES
French Aerospatial Agency

- Implemented a **user feedback system** on a machine learning model for the **detection of anomalies** in military satellite operations.
- Improved the **AI model** deployed on **operational satellite missions**.

May 2021 – Aug. 2021

Data Science intern — Marseille, France
KORNER (KLANIK)
Consulting in data

- Construction of a **neural network** to detect **Asian hornets and bees** through a **video stream**.
- The **AI model** is now used to build **Asian hornet traps** in France.

EDUCATION

2022 - 2025

PhD Degree — Palaiseau, France
Institut Polytechnique de Paris
Télécom Paris

Worked on deep learning methods to improve SAR satellite images, especially regarding data-fusion (optical and SAR) and interferometry.

2016 - 2022

Engineering - Master Degree — Lille, France
JUNIA
ISEN Lille

Master specialized in AI technology. Music and technology option.